

Kai Constantine

Software Engineer - London, UK

Education

BSc (Hons) Cyber Security - WMG, University of Warwick

September 2021 – July 2024

Achieved Upper Second Class (2:1)

A-Levels - Hurstpierpoint College

September 2019 – July 2021

A*, A*, A, B - Geography, Business Studies, Computer Science, EPQ

Experience

Cyber Attack and Defence Internship - Forvis Mazars LLP

July 2023 – August 2023

- Collaborated on four penetration testing projects, including IoT firmware analysis and malware evasion in Go.
- Wrote TDD-style unit tests and integrated automated checks into Git-based reviews, reducing regressions.
- Produced internal tooling and documentation; presented findings to stakeholders.
- Worked within an Agile Scrum team, contributing to sprint planning, stand-ups, and reviews.

Cyber Security Programming Tutor - University of Warwick

October 2023 – July 2024

- Delivered 10+ interactive programming tutorials to undergraduate and postgraduate students.
- Facilitated hands-on learning of algorithms and data structures, improving engagement and comprehension.

Projects

MoviePad — *React, Tailwind, Redux, FastAPI, Express, PostgreSQL, MongoDB, JWT, Docker*

Github: <https://github.com/Kaialogen/Movie-Pad>

- Led frontend development using React, Tailwind CSS, and Redux to build an intuitive interface for browsing, renting, and managing movies.
- Built dynamic user flows including catalogues, individual movie pages, shopping cart, and checkout.
- Implemented JWT authentication with refresh tokens, protected routes, and secure password hashing.
- Developed backend as two services (FastAPI for catalogue, Express for authentication) with clear contracts and typed DTOs.
- Designed PostgreSQL schemas for users and auth, and MongoDB models for catalogue data; added indexing and pagination for performance.
- Containerised services using Docker Compose with environment-driven configuration and health checks.

Hurst Chat — *Node.js, Express, PostgreSQL, Docker, JWT, React, Tailwind*

Github: <https://github.com/Kaialogen/Hurst-Chat>

- Designed and implemented RESTful APIs for user authentication, messaging, and session management.
- Integrated JWT-based authentication and secure password hashing.
- Modelled relational data using PostgreSQL with enforced referential integrity.
- Built a responsive React frontend enabling real-time messaging and dynamic interaction.

Technical Skills

Programming Languages	Python, TypeScript, Go, HTML/CSS, SQL
Frameworks & Libraries	Express, FastAPI, React, Next.js, Tailwind CSS
Databases	PostgreSQL, MongoDB, SQLite
Development Tools	Docker, Git, Linux, AWS, Jest, Pytest
Python Libraries	Numpy, Pandas